

Draft 6 — Author Action Checklist

The Face-Adjacency Graph of the Truncated Octahedron

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Recommendation: Accept with minor revisions. No confirmed critical or major mathematical error in the examined claims. Five groups of Draft 5 corrections are substantively addressed.

Before final release

1. **Finalize the immutable verification deposit (Section 8, pp. 10–11).** Supply the submitted Zenodo DOI or complete pinned commit; ensure the code, CSV, generated figure, certificate, dependency requirements and reviewer checks match the submitted PDF. Re-run the complete archive workflow and retain its output. In the public plotting script, correct the remaining “Draft 5” identification in the header.
2. **Streamline historical attribution (Acknowledgements, pp. 11–12).** Preserve the documented scholarly contributions while shortening per-round operational history; avoid implying independently conducted human refereeing or endorsement where the record only supports successive AI-assisted review rounds. Confirm authorization before naming a third party’s internal review method.

Optional improvements

1. Give readers a one-sentence guide to the difference between hand proof, exact finite computation, numerical comparison, and proof-assistant validation, referencing the existing Table 2.
2. In Section 6, emphasize near the figure reference that interpolating line segments join sorted eigenvalues and are not continuous analytic eigenbranches.

Mathematical findings retained

The ordinary characteristic polynomial, the $q = 6$ and $q = 12$ magnetic characteristic polynomials, the third- and fourth-moment identities, the spanning-tree count 101,154,816, and the five Fedorov comparison polynomials were corroborated. The absence of a Lean check is declared correctly; this does not itself constitute an error in a hand-proved mathematical note.

The full scientific reasoning, supporting identities, scoring, and the bounded assessment scope are provided in the accompanying referee report.